

1 Theoretical Foundation

The exercise consists of determining the position of objects in space, depending on the distance between them.

Considering a matrix:

$$\mathbf{D} = \begin{pmatrix} 0 & d_{12} & \dots & d_{1n} \\ d_{21} & 0 & \dots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \dots & 0 \end{pmatrix} \quad (1)$$

which represents the pairwise distance between the objects. The Matrix is symmetric.

The objective is to determine the position of the objects, in this case 2 dimensional, represented with the matrix:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}. \quad (2)$$

1.1 Optimization Problem

The objective is to find $\mathbf{X}$, so that the the distance between the positions is approximatly the value of the $\mathbf{D}$ matrix.

Meaning, the objective is:

$$\min_{\mathbf{X}} h_{\mathbf{X}}(\mathbf{D}) = \sum_{i=2}^n \sum_{j=1}^{i-1} |d_{ij} - \|x_i - x_j\|| \quad (3)$$

Since the absolute of a value is not a differentiable value, the l2-Norm is used.

$$\min_{\mathbf{X}} h_{\mathbf{X}}(\mathbf{D}) = \sqrt{2 \sum_{i=2}^n \sum_{j=1}^{i-1} (d_{ij} - \|x_i - x_j\|)^2} \quad (4)$$